Style Guide

1. Character Spacing

- Add a space before and after operators
- No spaces before or after array operators
- No space between a function name and its parameters

2. Blank Lines

- Between functions
- Between the file header and the rest of the file
- Between groups of related statements, where it helps readability

3. Indentation

- **C++:** use 4 spaces per indent
- **Java:** use 2 spaces per indent
- Indent in for loops and while loops even if not required in the coding language
- Indent conditionals
- Indent within function declarations
- Long argument lists may wrap onto the next line, indented 2 more spaces (Java)

4. Variable Declaration Placement and Alignment

- Using C++11 auto type deduction for longer types (C++ only)
- Also using templates for repeated use of functions with different types (C++ only)
- One variable declaration per line, except closely related variables of the same type (for example JLabel subtotalLabel, taxLabel;)
- Declare variables at the beginning of a function in C++, and near first use in Java
- Use camelCase for variable and method names
- Short single statements may share a line (for example rs.close(); ps.close();)

5. Case Statement Alignment

- Follow the rules in Section 3
- Opening braces go inline with the control structure declaration
- else and else if go on the same line as the closing brace unless short functions
- An if or else body may omit braces only if the whole statement fits on one line or if it is short enough to make sense to the reader

6. Comment Layout

- Place comments before what they describe, or inline if short enough
- Put a comment at the top of each file explaining its purpose
- Put a comment before each function or method explaining what it does
- In Java GUI files, major sections may be marked with // ---- Section name ----
- Include a README that describes the purpose of each file

7. Additional Attributes

- **Constants:** all caps with underscores (TAX_RATE)
- **Variables:** first word lowercase, then each new word capitalized, with no underscores (orderLines, inCart)
- **Functions and methods:** same as variables (showCategory)
- **Classes:** every word capitalized, with no underscores (CashierGUI, InventoryDAO)
- **File names:** all lowercase in C++ with underscores. Java files must match their class name (CashierGUI.java)
- **Variable names should describe what they hold.** Short names (b, c, m, p, it, ps, rs, e) are allowed for local variables used inside a single method
- **Allowed abbreviations:** conn, db, dao, qty, id, sub, tot

8. Style Guide Enforcement

We will use a linter to strictly enforce these guidelines, with no exceptions. Anyone who doesn't follow the rules will be temporarily suspended from committing new code.